

Project CAD / Drawing Structure Document

HYDERABAD HYPERSCALE DATA CENTRE - PHASE 1 TYPICAL DATA HALL LAYOUT - SINGLE LINE DIAGRAM & FLOOR PLAN PROJECT CAD / DRAWING STRUCTURE DOCUMENT

Document No.	HYD-DC-P1-DWG-REG-001
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Prepared By	BIM / Design Coordination Team
Classification	Controlled Document - Engineering Distribution Only
Approved By	Lead Design Coordinator

1. Purpose

This document defines the CAD/drawing file structure, numbering convention, and current revision/approval status of all drawings associated with the Typical Data Hall Layout package, comprising the Power Single Line Diagram, Cooling Layout, Rack Elevation, and Cable Tray Routing drawing sets for Hyderabad Hyperscale Data Centre - Phase 1. It serves as the master drawing register referenced by all site, procurement, and commissioning teams.

2. Drawing Numbering Convention

Drawing numbers follow the format: **HYD-DC-P1-[DISCIPLINE]-[HALL]-[SEQ]-[REV]**

Segment	Description	Example
DISCIPLINE	2-3 letter discipline code (see Section 3)	ELE, MEC, CAB, ARC
HALL	Data hall reference or GEN for general/typical	DH1, DH2, DH3, DH4, GEN
SEQ	Sequential drawing number within discipline	001, 002 ...
REV	Revision letter	A, B, C ...

Example: HYD-DC-P1-ELE-GEN-001-C = Electrical discipline, general/typical (not hall-specific), drawing 001, Revision C.

3. Discipline Codes

Code	Discipline
ARC	Architectural
STR	Structural
ELE	Electrical / Power
MEC	Mechanical / Cooling
CAB	Structured Cabling / Cable Tray
FIR	Fire Protection
SEC	Security
BMS	Building Management / Controls

4. CAD File Descriptions

4.1 Power Single Line Diagram (SLD)

File Series: HYD-DC-P1-ELE-GEN-0xx

Depicts the complete electrical distribution hierarchy from 132 kV utility intake through step-down transformers, HT/MV switchgear, DG synchronization panels, UPS input/output boards, busway distribution, and down to rack-level iPDU A/B feeds. Includes protective device ratings, breaker sizes, cable/busway ampacity annotations, and 2N train separation clearly color-coded (Train A in blue, Train B in red per drawing legend).

Native Format: AutoCAD Electrical (.dwg), exported reference set in .pdf and .dwf **Coordination Model:** Linked to Revit MEP federated model (Level 400 detail)

4.2 Cooling Layout

File Series: HYD-DC-P1-MEC-GEN-0xx / HYD-DC-P1-MEC-DH[n]-0xx

Shows chilled water plant room layout, primary/secondary loop piping schematic, CRAH unit placement per hall zone, hot-aisle containment boundaries, D2C CDU placement in high-density GPU zones, and leak detection rope routing under liquid-cooled racks.

Native Format: AutoCAD MEP (.dwg), Revit MEP model (.rvt) for hall-specific sheets **Coordination Model:** Federated with structural and electrical for clash detection (LOD 400)

4.3 Rack Elevation

File Series: HYD-DC-P1-ELE-DH[n]-1xx

Elevation drawings showing standard 42U EIA rack and OCP ORv3 rack configurations, iPDU A/B mounting positions, cable management arms, blanking panel requirements, and rack numbering labels per hall row.

Native Format: AutoCAD (.dwg), typical elevation templates reused across halls with hall-specific row/position labeling

4.4 Cable Tray Routing

File Series: HYD-DC-P1-CAB-GEN-0xx / HYD-DC-P1-CAB-DH[n]-0xx

Overhead and under-floor tray routing for structured cabling (OS2 singlemode backbone, OM4 horizontal, Cat 6A management), showing tray fill ratios, power/data pathway separation (minimum

300mm parallel), and dedicated InfiniBand/RoCEv2 basket tray routing in GPU-dense zones with bend-radius compliance callouts.

Native Format: AutoCAD (.dwg), Revit MEP model for federated clash detection

5. Master Drawing Register - Status Summary

Drawing No.	Title	Discipline	Revision	Status	Approved By	Approval Date
HYD-DC-P1-ELE-GEN-001-C	Power Single Line Diagram - Utility to UPS Input	ELE	C	Issued for Construction	Lead Electrical Engineer	02 Jun 2026
HYD-DC-P1-ELE-GEN-002-C	Power Single Line Diagram - UPS Output to Rack (2N)	ELE	C	Issued for Construction	Lead Electrical Engineer	02 Jun 2026
HYD-DC-P1-ELE-GEN-003-B	DG Synchronization & Paralleling Schematic	ELE	B	Under Review	-	-
HYD-DC-P1-MEC-GEN-001-C	Chilled Water Plant Room Layout	MEC	C	Issued for Construction	Lead Mechanical Engineer	28 May 2026
HYD-DC-P1-MEC-DH1-002-B	Data Hall 1 - CRAH & Containment Layout	MEC	B	Issued for Construction	Lead Mechanical Engineer	15 Jun 2026
HYD-DC-P1-MEC-DH2-002-B	Data Hall 2 - CRAH & Containment Layout	MEC	B	Issued for Construction	Lead Mechanical Engineer	15 Jun 2026
HYD-DC-P1-MEC-DH3-002-A	Data Hall 3 - CRAH & Containment Layout	MEC	A	Under Review	-	-
HYD-DC-P1-MEC-DH4-002-A	Data Hall 4 - CRAH & Containment Layout	MEC	A	Under Review	-	-

Drawing No.	Title	Discipline	Revision	Status	Approved By	Approval Date
HYD-DC-P1-MEC-GEN-010-B	D2C Liquid Cooling / CDU Placement – GPU Zone	MEC	B	Issued for Construction	Lead Mechanical Engineer	20 Jun 2026
HYD-DC-P1-ELE-DH1-101-B	Rack Elevation – Standard 42U (Typical)	ELE	B	Issued for Construction	Lead Electrical Engineer	10 Jun 2026
HYD-DC-P1-ELE-DH1-102-A	Rack Elevation – OCP ORv3 High-Density Zone	ELE	A	Under Review	–	–
HYD-DC-P1-CAB-GEN-001-C	Cable Tray Routing – Backbone (MDA to HDA)	CAB	C	Issued for Construction	BIM Coordination Lead	05 Jun 2026
HYD-DC-P1-CAB-GEN-002-B	Cable Tray Routing – Infini-Band/RoCEv2 GPU Fabric	CAB	B	Issued for Construction	BIM Coordination Lead	18 Jun 2026
HYD-DC-P1-CAB-DH3-003-A	Data Hall 3 – Horizontal Cabling & ZDA Layout	CAB	A	Draft – Not Issued	–	–
HYD-DC-P1-FIR-GEN-001-B	Fire Suppression Layout – VESDA & Clean Agent Zones	FIR	B	Issued for Construction	Fire & Life Safety Engineer	22 May 2026
HYD-DC-P1-SEC-GEN-001-B	Physical Security Layout – Access Control & CCTV	SEC	B	Issued for Construction	Security Design Lead	22 May 2026

Legend: *Draft – Not Issued* = internal working draft; *Under Review* = submitted for discipline lead sign-off; *Issued for Construction (IFC)* = approved and released to site.

6. Revision Control Notes

- All drawings are maintained in the central Common Data Environment (CDE) with full version history; superseded revisions are archived, not deleted.
- Any drawing marked “Under Review” or “Draft – Not Issued” **must not** be used for procurement or site installation purposes.
- Revision changes affecting interfacing disciplines (e.g., an ELE revision impacting CAB tray routing) trigger a mandatory cross-discipline clash-check re-run before re-issue.
- Federated BIM model (LOD 400/500 as applicable per Section 4) is the single source of truth for clash detection; individual 2D sheet drawings are derived outputs and must match the federated model at time of issue.

End of Document – HYD-DC-P1-DWG-REG-001, Version 1.4